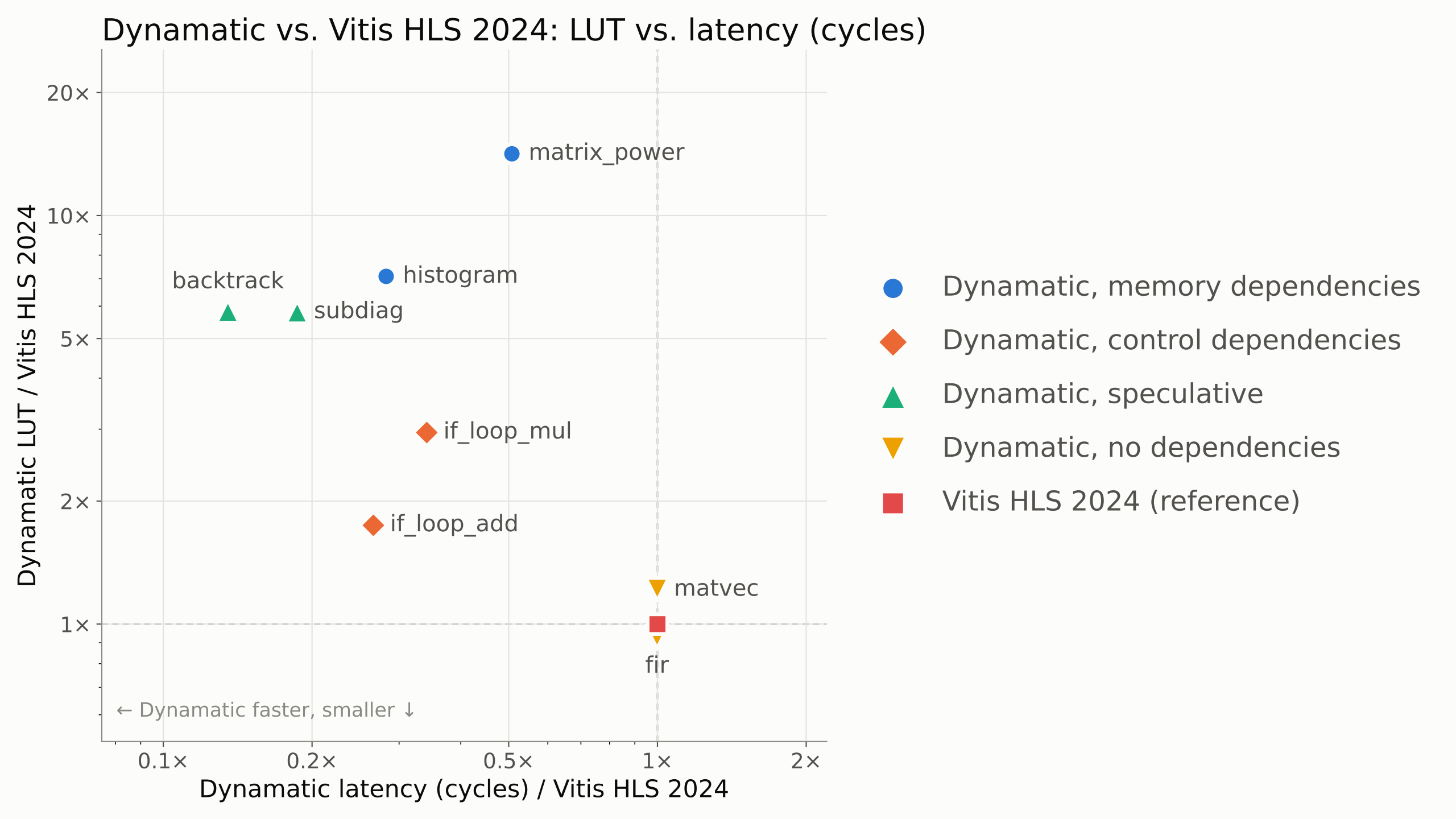
Dynamatic vs. Vitis HLS 2024: LUT vs. latency (cycles)
Dynamatic LUT / Vitis HLS 2024
20×
10×
5×
2×
1×
matrix_power
histogram
backtrack
subdiag
if_loop_mul
if_loop_add
matvec
fir
← Dynamatic faster, smaller ↓
0.1×
0.2×
0.5×
1×
2×
Dynamatic latency (cycles) / Vitis HLS 2024
Dynamatic, memory dependencies
Dynamatic, control dependencies
Dynamatic, speculative
Dynamatic, no dependencies
Vitis HLS 2024 (reference)